

Mahriar Gharaghani

Research-Oriented Curriculum Vitae

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Location	Tehran, Iran
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Research Areas	Artificial Intelligence, Reinforcement Learning, Decision Intelligence, Operations Research-inspired AI Systems

Research Profile

Research-oriented AI engineer with academic and practical experience at the intersection of artificial intelligence, sequential decision-making, and project-oriented resource allocation. Current work focuses on the design of intelligent decision-support systems for project portfolio budgeting under uncertainty and dynamic cash-flow constraints, combining simulation, optimization, and reinforcement learning. Prior professional experience in project control and project management has provided the applied context that motivates this research direction.

Research Interests

Core Interests	Reinforcement learning, sequential decision-making under uncertainty, optimization-informed AI, simulation-based evaluation, decision-support systems
Application Domains	Project portfolio management, budget allocation, cash-flow constrained planning, industrial and organizational decision systems
Related Technical Interests	Retrieval-augmented generation (RAG), LLM-based workflow systems, ML systems engineering, AI-assisted knowledge tools

Education

Degree	Institution	Dates	Details
M.Sc. in Engineering Management	Iran University of Science and Technology (IUST), Tehran, Iran	2023–2026	In progress. Thesis: <i>Autonomous E</i>
B.Sc. in Industrial Engineering	Isfahan University of Technology (IUT), Isfahan, Iran	2017–2023	Completed.
Ph.D. Entrance Exam (Computer Engineering — AI & Robotics)	National Universities of Iran	2026	Results pending.

Research Experience

Role / Context	Dates	Description
M.Sc. Thesis Researcher, IUST	2023–2026	Conducting graduate research on AI-based project portfolio budgeting
Industry-Supported Research Context	2023–2026	Thesis work supported under a

Selected Research and Technical Projects

Project	Period	Description
Autonomous Budgeting AI Agent	2023– present	Developed a research-oriented system for sequential budget allocation
RAG4Konkur	2025–2026	Developed a local retrieval-augmented study assistant for flashcards
End-to-End House Price ML Pipeline	2024	Implemented a full machine learning workflow for house price prediction
PyTorch CIFAR-10 Pipeline	2024	Implemented an image-classification pipeline using PyTorch
Model Serving with FastAPI and Docker	2024	Designed a deployable inference workflow for machine learning models
Robotic Arm Reinforcement Learning Simulation	2024	Explored reinforcement learning in a robotic arm control simulation
Decision-Support and Visualization Tools	2023– present	Developed several applied prototypes including logistics heatmap tool

Professional Experience

Position	Dates	Description
Project Management Officer, Iran Energy Industries Corporation (IEIC)	2023	Worked in a project-management environment with exposure to organizational strategy
Project Control Specialist, Jahanpars Group	2019–2020	Contributed to project control activities while still an undergraduate student
Independent Technical Reorientation	2023– present	Undertook a deliberate transition from management-oriented roles to technical research

Technical Preparation and Independent Study

Software Engineering	Completed substantial self-directed study in web and software engineering, including responsive web design, JavaScript algorithms and data structures, front-end libraries, data visualization, relational databases, and back-end APIs.
ML Systems Preparation	Worked on model development pipelines, deployment workflows, API serving, containerization, and experiment-tracking concepts relevant to practical ML engineering.
AI Application Development	Explored LangChain-based systems, RAG pipelines, workflow automation, and local AI tool development for domain-specific use cases.

Technical Skills

Programming Languages	Python; JavaScript; TypeScript
Scientific / Technical Writing	L ^A T _E X; Markdown
Backend and APIs	FastAPI; Django; Express.js
Frontend Development	React.js
Machine Learning / AI	Machine learning; deep learning workflows; reinforcement learning exposure; prompt engineering; LangChain; retrieval-augmented generation (RAG)
Data Work	Exploratory data analysis; preprocessing; feature engineering; data visualization
Deployment / Systems	Docker; model serving; deployment concepts; experiment tracking concepts
Workflow / Automation	N8N-style automation workflows; AI-assisted process tooling

Certifications and Structured Training

freeCodeCamp (selected tracks)	Responsive Web Design; JavaScript Algorithms and Data Structures; Front End Development Libraries; Data Visualization; Relational Database; Back End Development and APIs
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Languages

Persian	Native
English	Near-native / professional proficiency